

Name of the Student WafarRoll No. 63

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF ELECTRICAL ENGINEERING
END TERM EXAMINATION NOVEMBER-2025

Course: **B.Tech**Subject Name: **EMEC-I**Time : **120 minutes**NOTE: **Answer all Questions**Semester **III**Branch: **Electrical Engg.**Subject Code: **EE-24212**Max Marks: **40**

Q.No.	Question	Marks	CO
1. <u>a</u>	A delta connected balanced 3-phase load is supplied from a 3-phase, 400 V supply. The line current is 20 A and the power taken by the load is 10 kW. Find (i) impedance in each branch (ii) line current, power factor and power consumed if the same load is connected in star.	05	CO1
<u>b</u>	Explain how three-phase power is measured using the two-wattmeter method. Draw the phasor diagram of the two-wattmeter method for a purely resistive load. Show mathematically that both wattmeters read equal power.	05	CO3
2. <u>a</u>	A Scott connected transformer system supplies two single phase transformers at 110 V, each drawing 250 kW. The load on the leading phase is at unity power factor and on the other phase is 0.8 power factor. The three phase input line voltage is 22000 volts. Find the line currents on the primary side. Neglect the magnetizing current and leakage impedance.	05	CO2
<u>b</u>	Assume a 20 kVA, 2000/200 V, 2-winding transformer having 98% efficiency on full load and unity power factor is reconstructed to obtain autotransformer. Show the connection and analyze for 2000/2200 V autotransformer.	05	CO3
3. <u>a</u>	A 600 kVA load at power factor of 0.7 lagging is supplied by three single phase transformers of 250 kVA, 2200/220 V connected in delta-delta connection. If one transformer is damaged and removed from the service, calculate for the V-V bank: (i) the kVA supplied by each transformer (ii) % load supplied by each transformer (iii) kVA rating of the transformer in V-V bank (iv) Ratio of V-V bank to delta-delta bank transformer rating and (v) % increase in load on each transformer when one transformer is removed.	05	CO2
<u>b</u>	Describe the effect of armature reaction on the main field flux distribution. How does it cause flux distortion? Explain with neat sketches.	05	CO1
4. <u>a</u>	A 500 V DC shunt motor runs at 250 rpm at rated full load condition and takes an armature current of 200 A. The armature resistance is 0.12 ohms. Find the speed of the motor when the field resistance is increased such that the flux is reduced to 80% of the normal value and the motor is loaded for an armature current of 100 A.	05	CO2
<u>b</u>	Explain the step-by-step process of voltage build-up in a DC shunt generator. What is the role of residual magnetism? Define critical field resistance and critical speed. Explain their significance in voltage build-up.	05	CO1